

EE P 564: TinyML

Week 7: TinyML for Motion Classification

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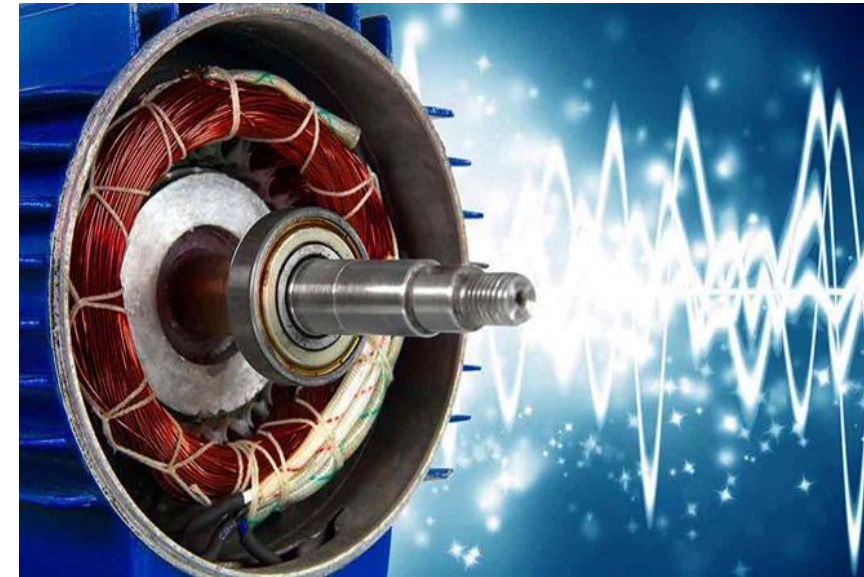
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Topics Covered

- Background on Motion Classification and different types
- Supervised learning for motion classification
- Sensor interface for Mechanical Stress in Transport
- Model Training, Evaluation Metrics, and Deployment
- Lab 7: Motion Classification Data Collection, Model Training, Model Compression, and Deployment using Edge Impulse



❖ Relevant reading from the Textbook [Warden and Situnayake; O'REILLY Publisher] is Chapter 17

Motion Classification

- Why should we classify motion?
- What are the application domains where motion classification can be useful?
- What are the input data types that can effectively help to classify motion data?
- How is motion classification similar or different to gesture recognition?

	Input Video Clip	Motion Type Predicted
<i>Cart-wheel</i>		<i>Projectile</i>
<i>Jogging</i>		<i>Linear</i>
<i>Diving</i>		<i>Projectile</i>
<i>Drinking</i>		<i>Local</i>
<i>Fencing</i>		<i>Random</i>

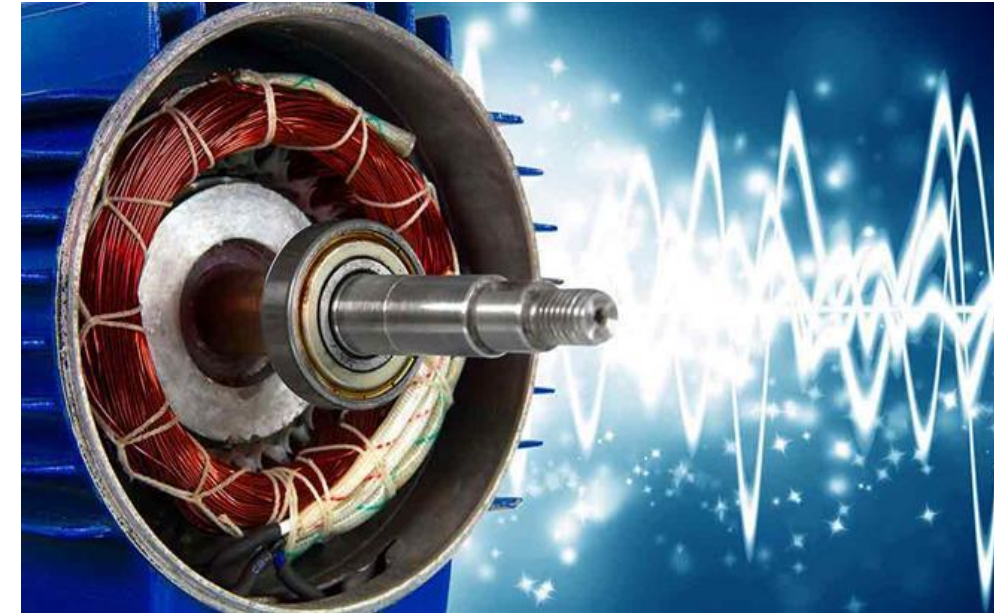
Classification of Different Data



Audio Classification



Image Classification



Motion Classification

What do we choose as input for motion classification?

Depends on the application!

Motion Classification

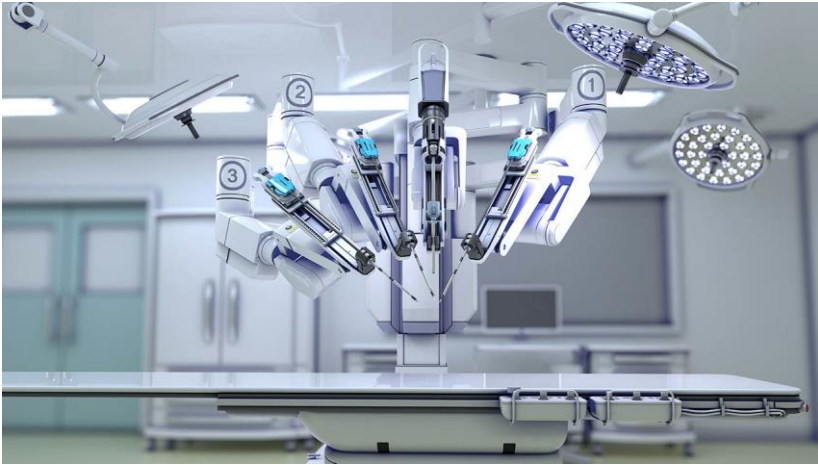


Classify Washing Machine cycles



Classify poses for games/ yoga/ martial arts

Applications of Motion Classification



Surgical Robotics

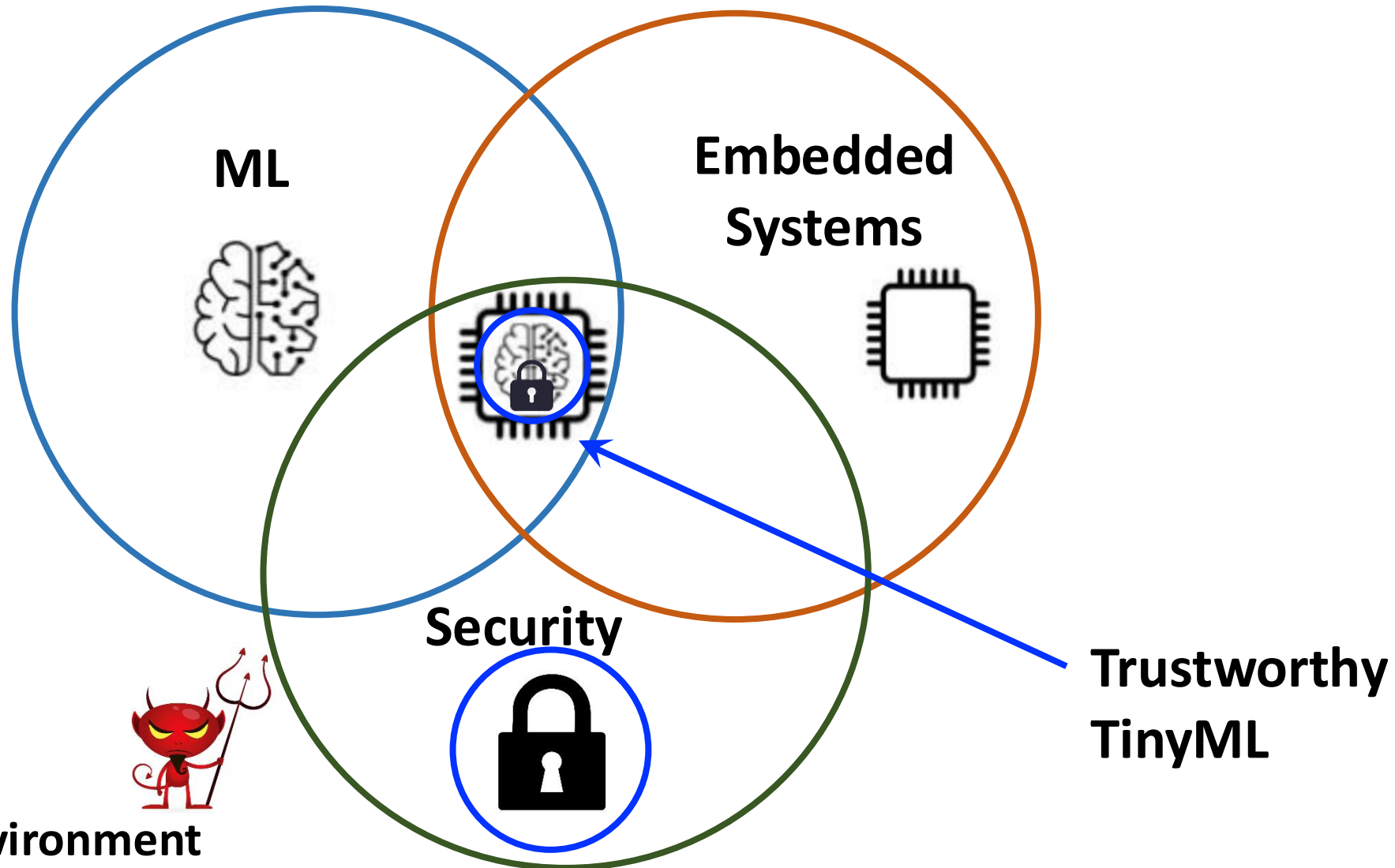


Transportation of Freight



Motion Capture Technology

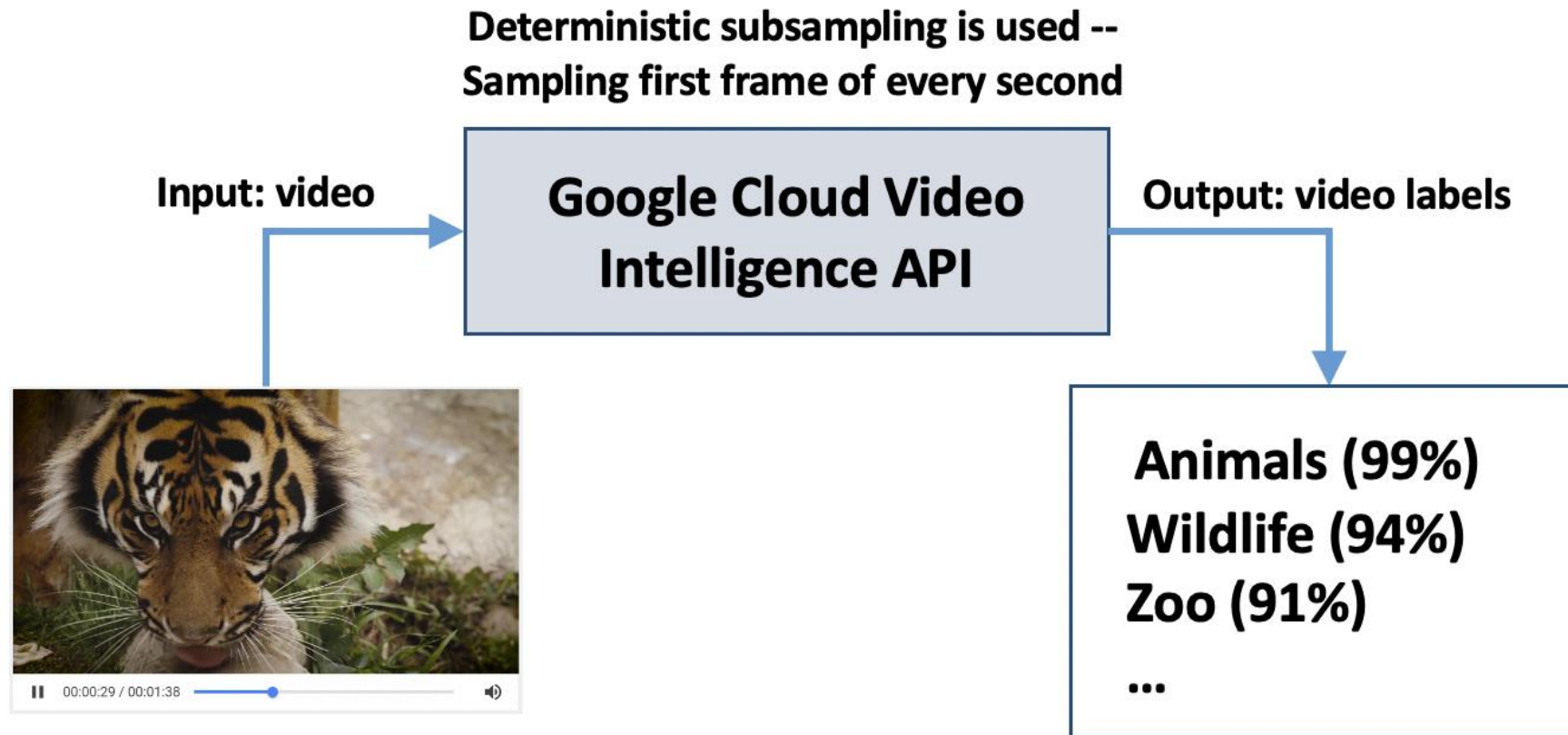
Can we attack motion classifier applications?



Untrusted Environment

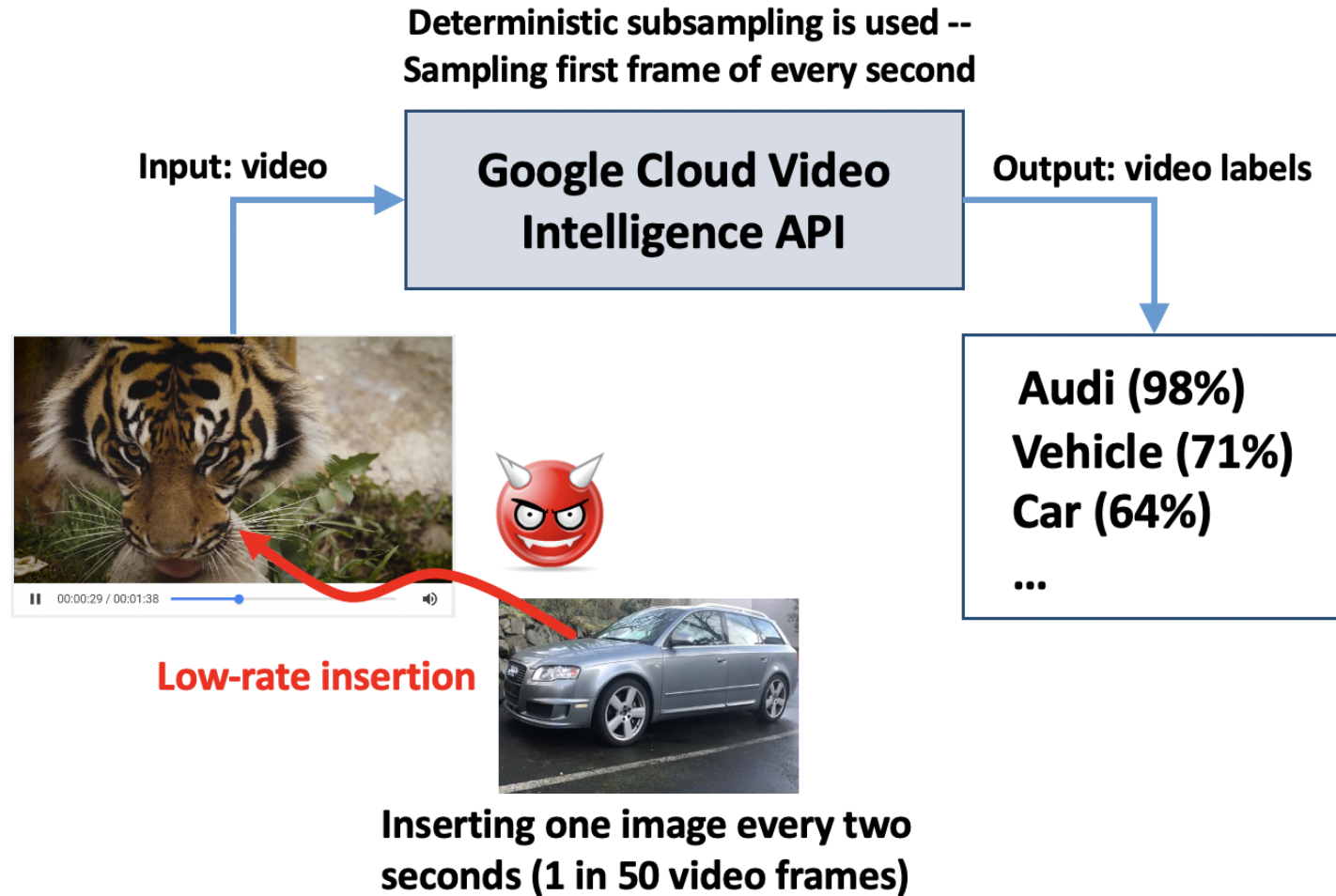


Google Video Analysis with DNN



Hosseini, H., Xiao, B. and Poovendran, R., "Deceiving Google's Cloud Video Intelligence API Built for Summarizing Videos." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2017.

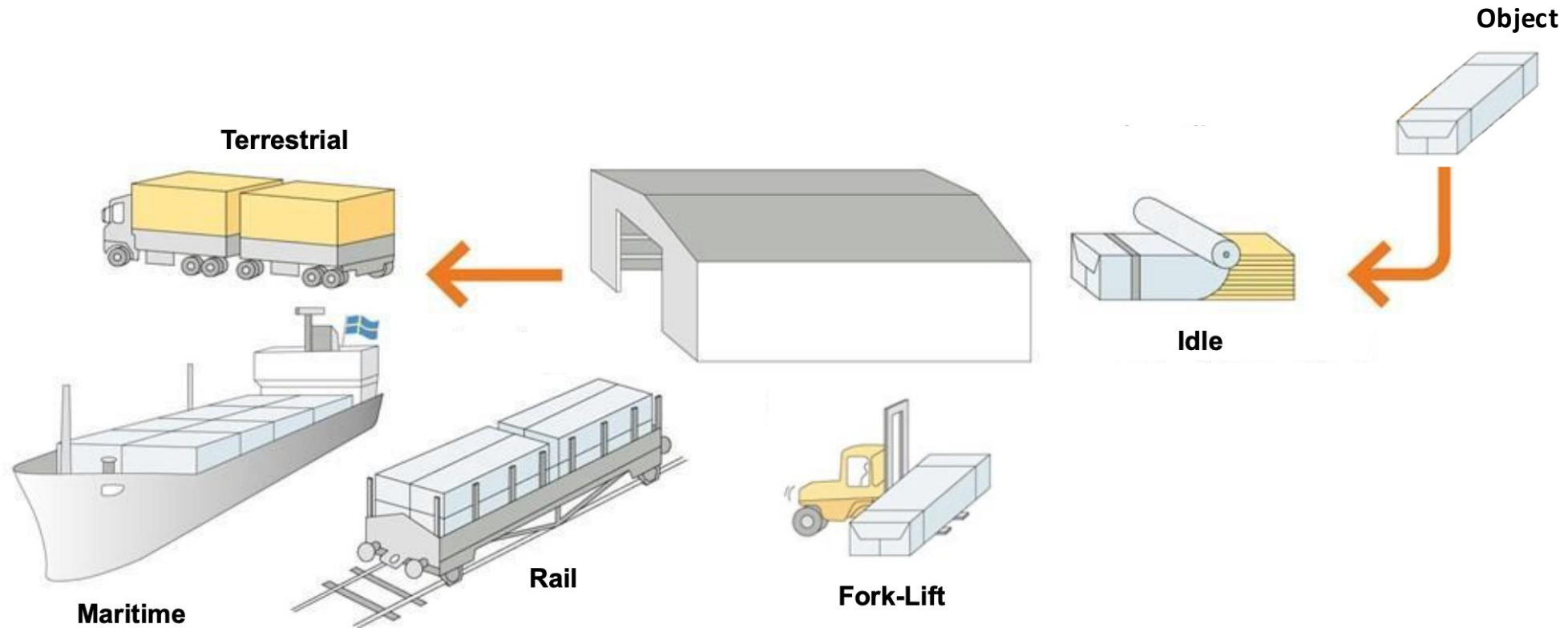
Decision Time Attack--With Domain Knowledge



Hosseini, H., Xiao, B. and Poovendran, R., "Deceiving Google's Cloud Video Intelligence API Built for Summarizing Videos." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*. 2017.

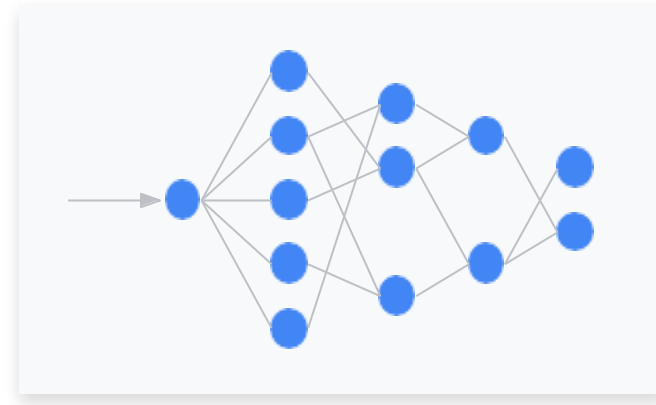
Motion Classification for Transport

- Motion classification for transport is essential to calculate the stress or force on objects in transport.
- Objects experience different types of stress based on the conditions of the route.



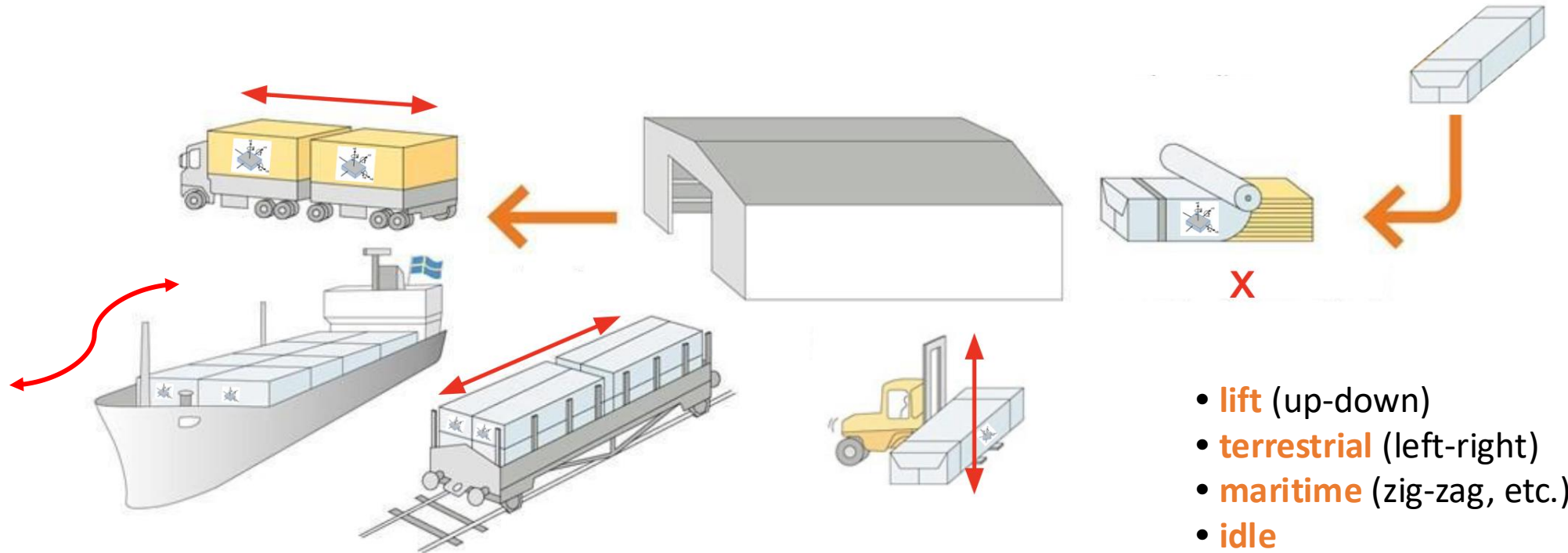
Challenges in implementation

- On-device Computing is necessary since streaming to the cloud might not be possible due to poor connection during freight transport.

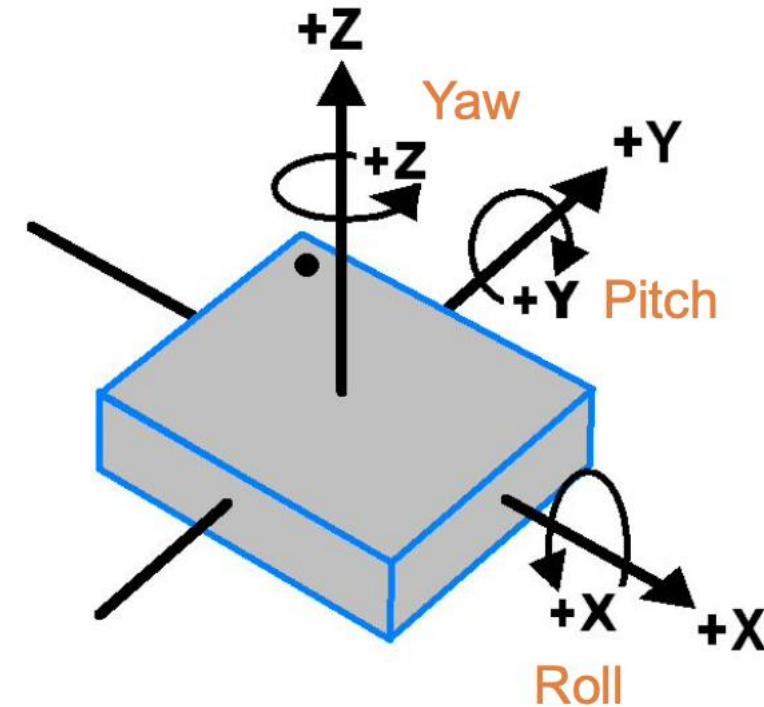
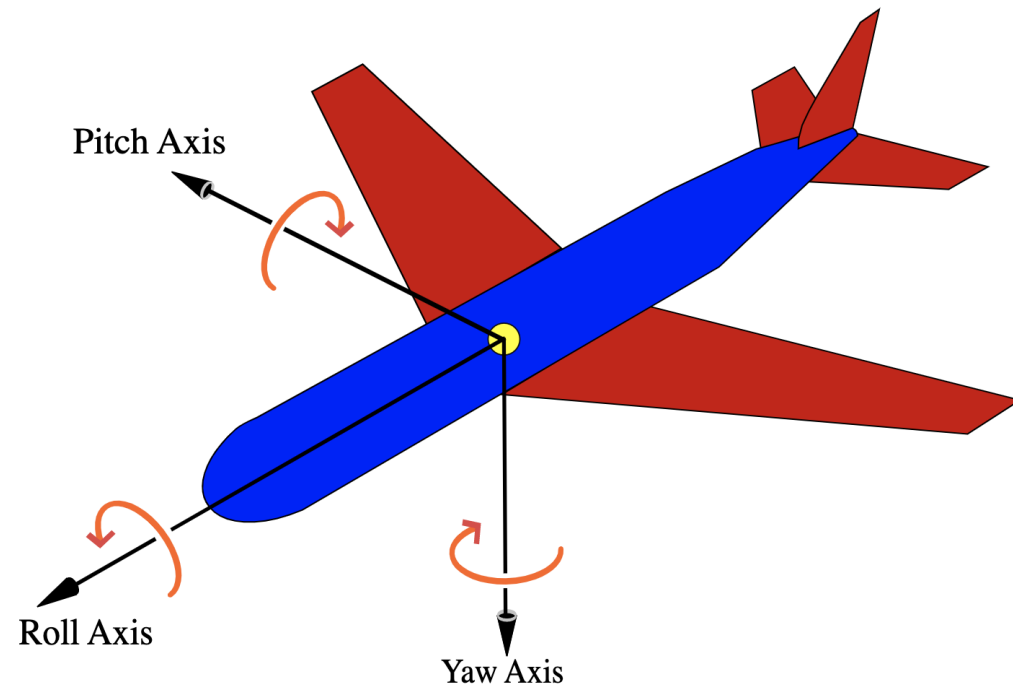


Degrees of Freedom

- Degrees of freedom (DoF) refer to the number of basic ways a rigid object can move through 3D space.
- We define various classes based on the transport type and the degree of freedom for object movement in each of the types.

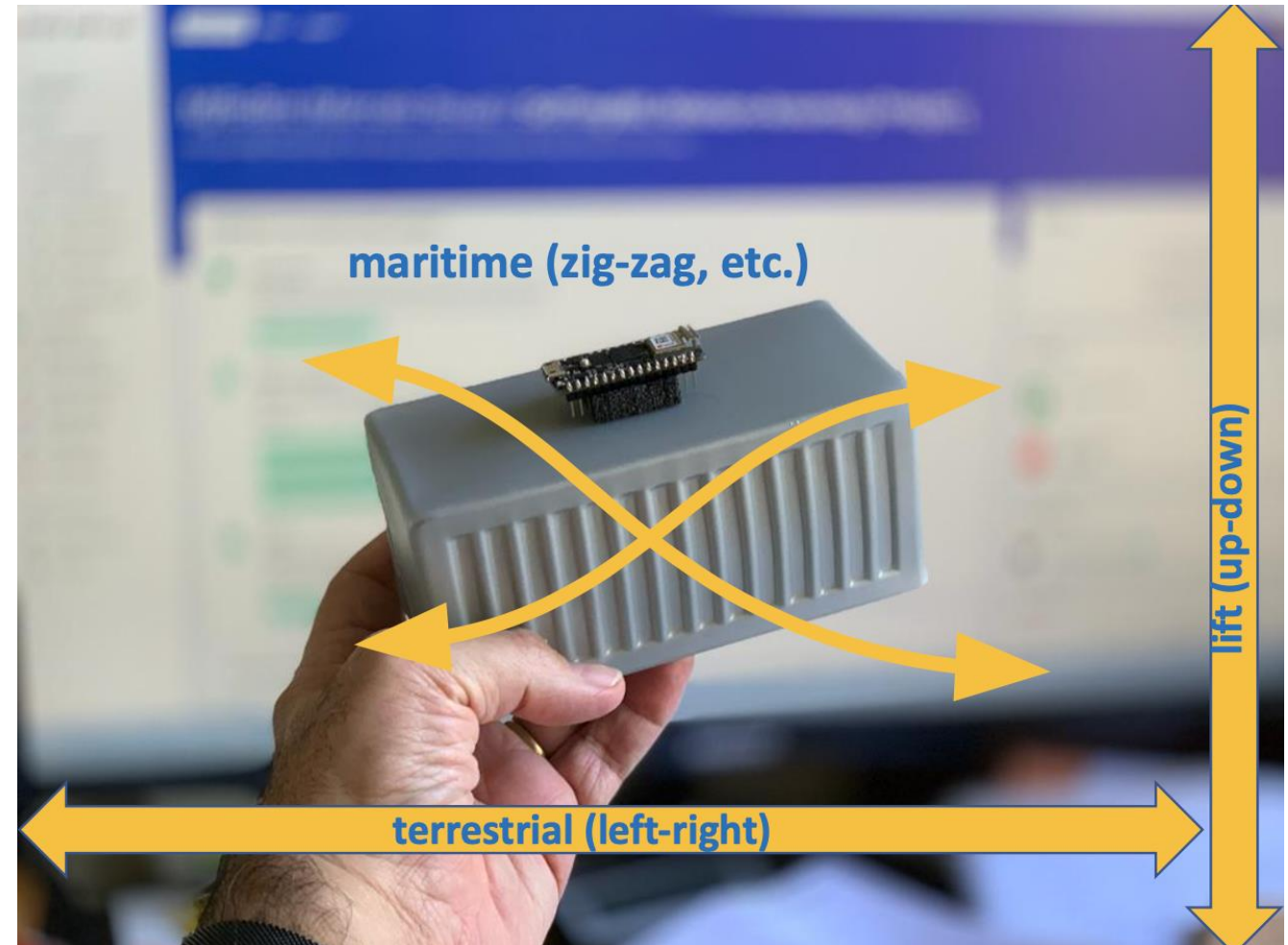
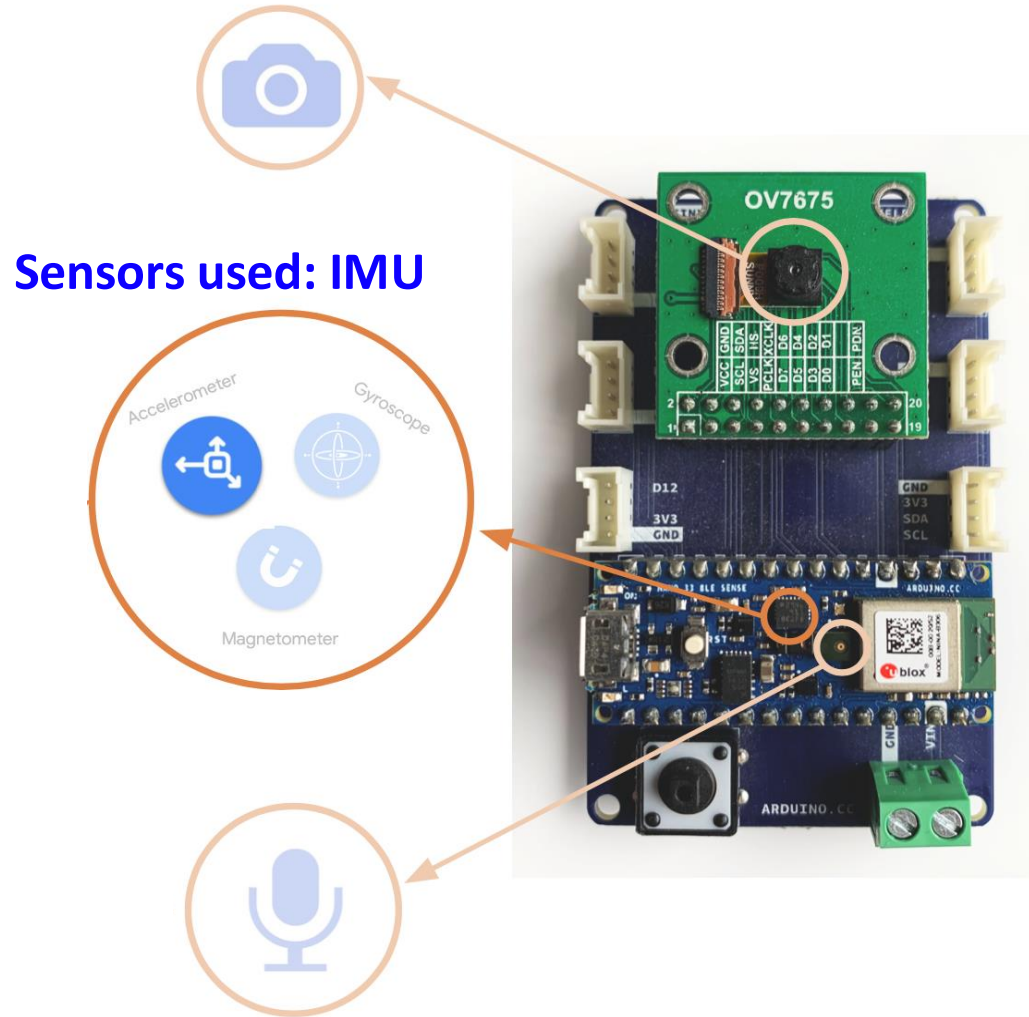


Inertial Measurement Units (IMU)



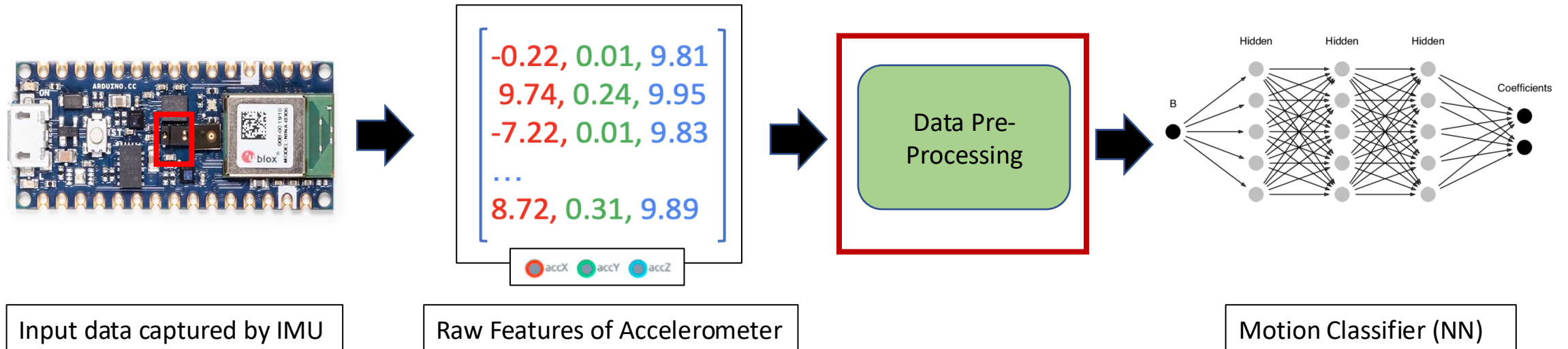
Yaw, Pitch and Roll measurements through Accelerometer and Gyroscope data

TinyML IMU Sensors



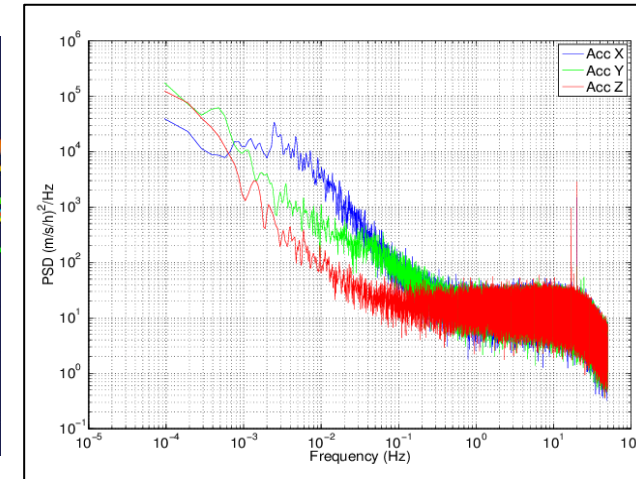
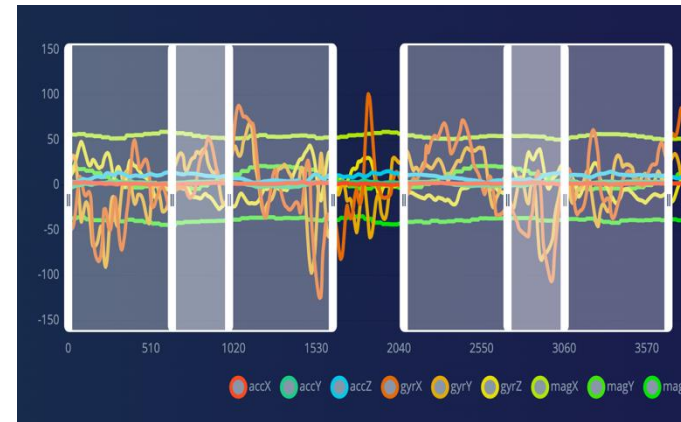
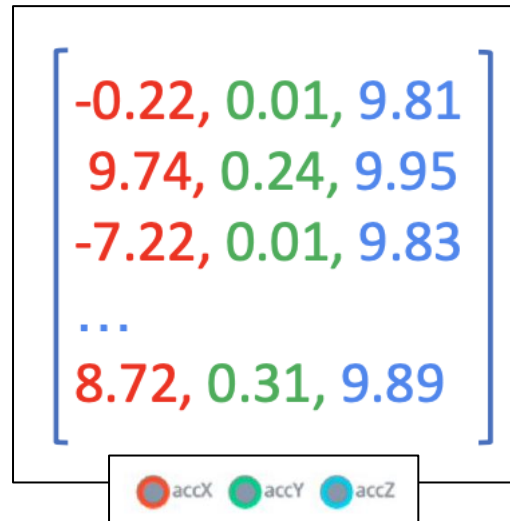
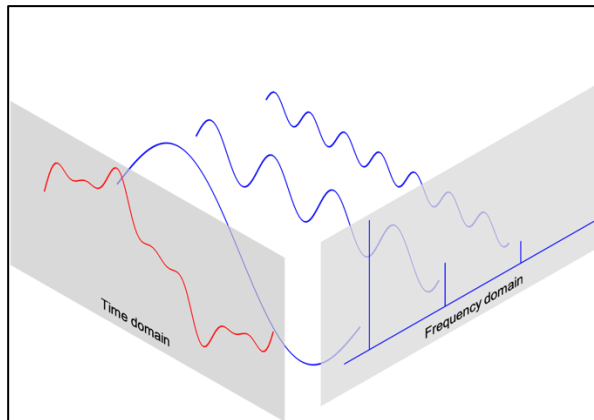
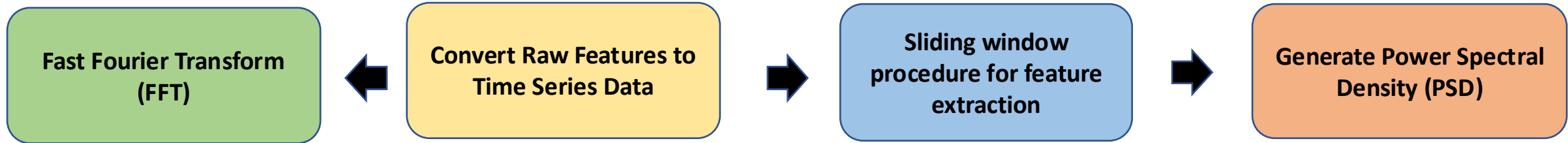
End-to-End flow of Motion Classifier

- End-to-End flow from capturing the motion data to classifying the different types of motion.
- We use a DNN Classifier as our model.



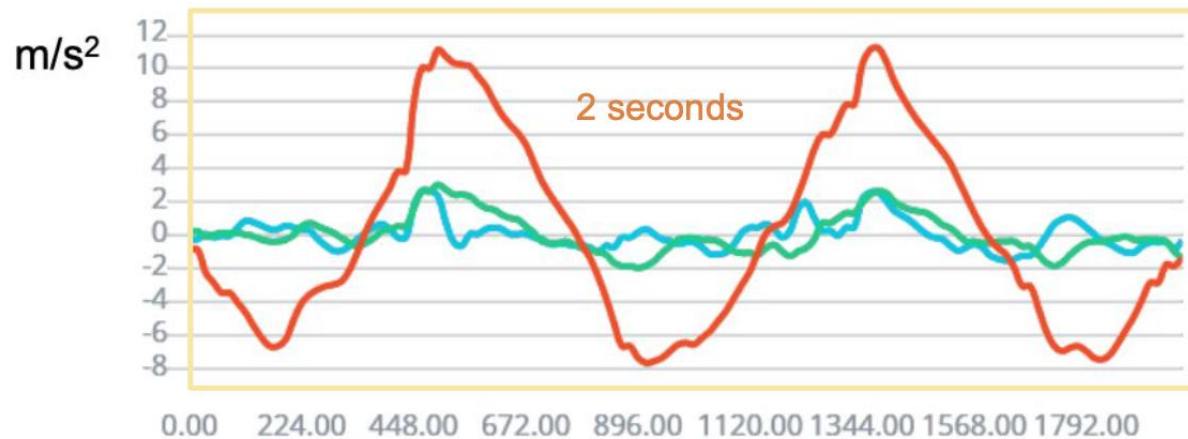
Data Pre-Processing

Data pre-processing steps:

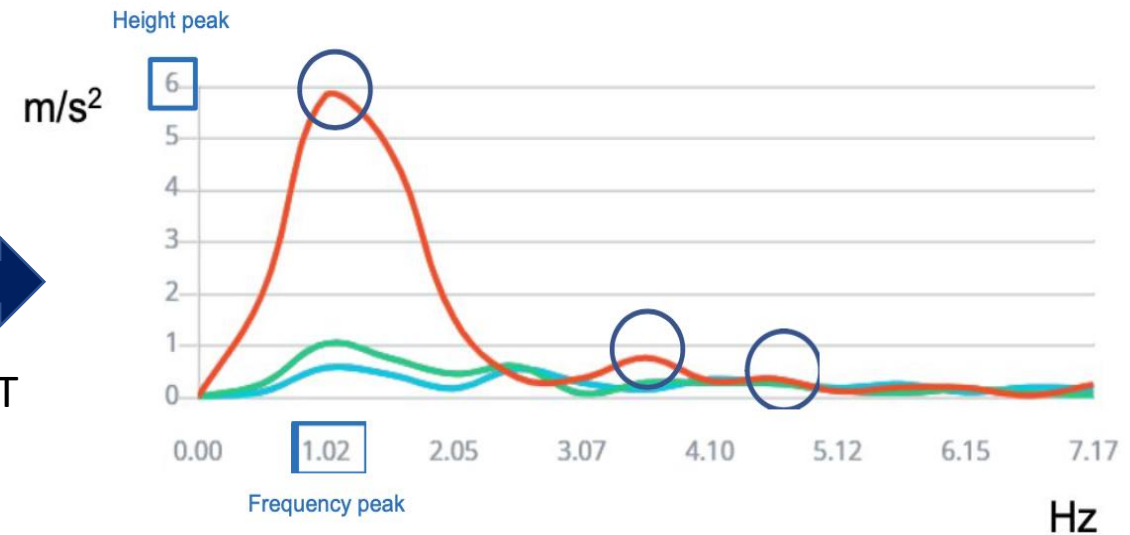


FFT for Accelerometer Data

- Calculating FFT for accelerometer data extracts **Height Peak** and **Frequency Peak** parameter from the raw data which can then be fed to the NN for gaining a compressed model to implement on tiny devices.



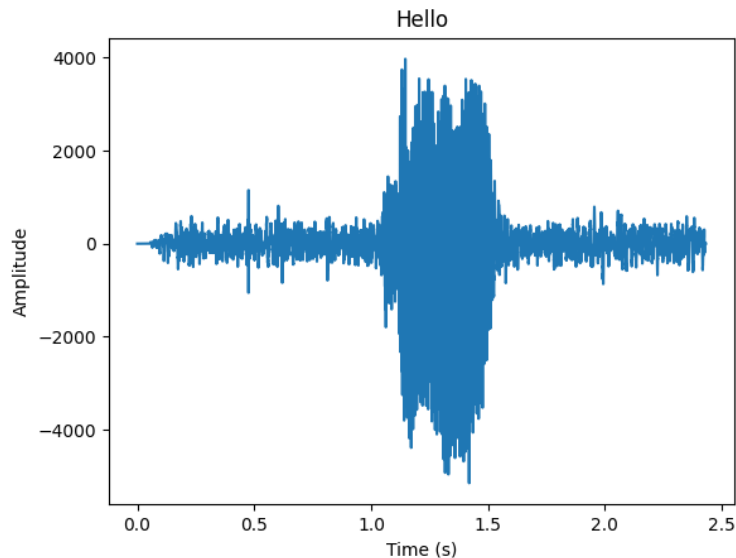
FFT



Recall.. Fast Fourier Transform (FFT)

- The time series signal (Time v. Amplitude) is transformed into the frequency domain (Frequency v. Amplitude) using Discrete Fourier transform.
- The Fast Fourier Transform (FFT) is an efficient algorithm to calculate the DFT of a sequence.

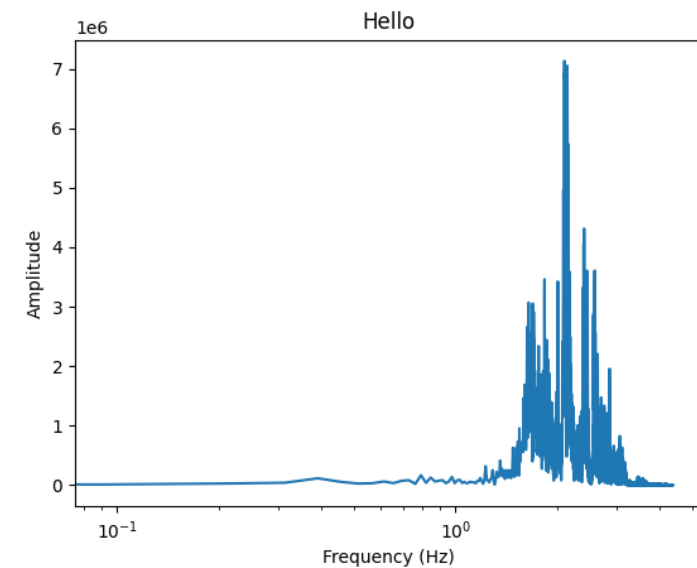
Word: Hello



Speech Signal in Time Domain

Fast Fourier Transform

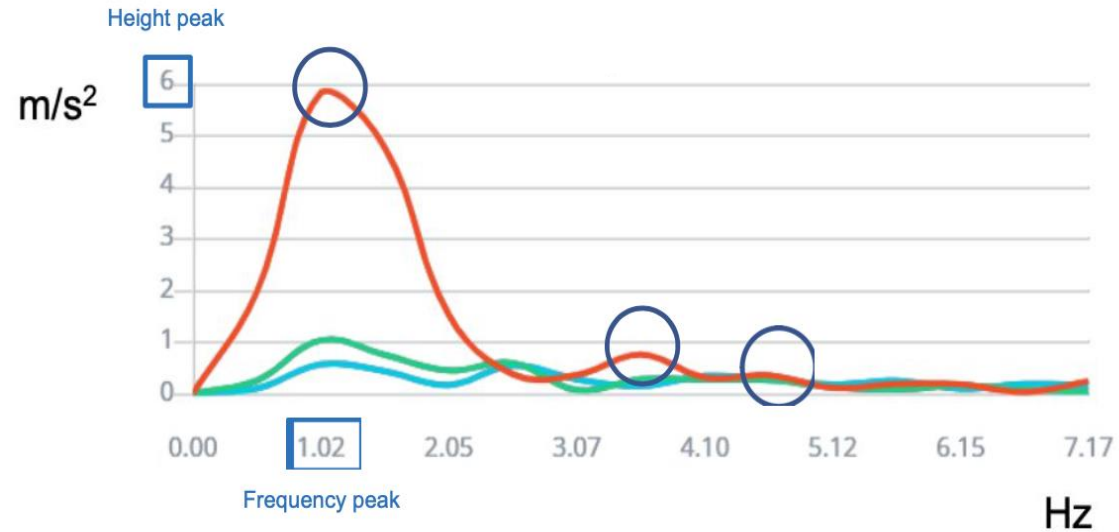
Word: Hello



Speech Signal in Frequency Domain

Power Spectral Density (PSD)

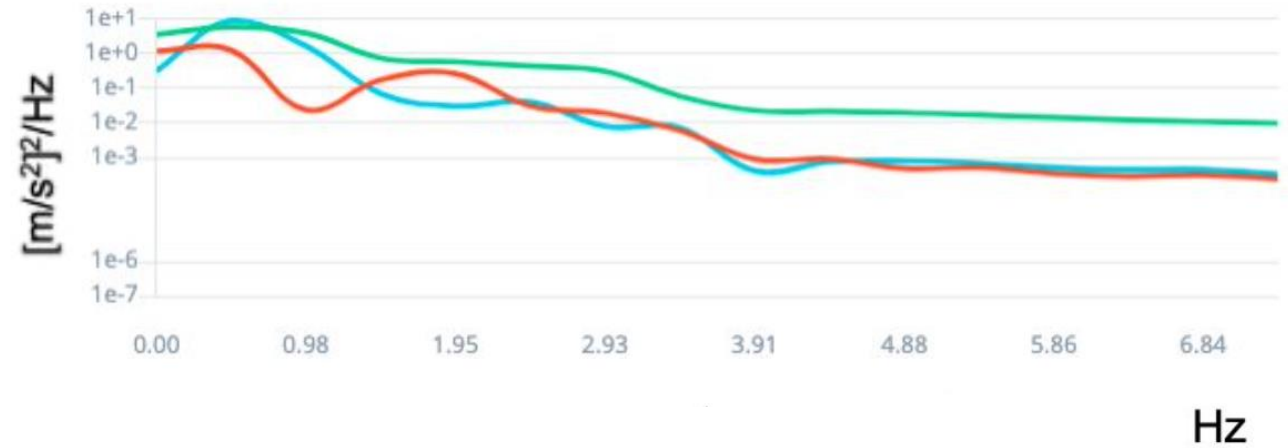
Calculating PSD from FFT of the signal:



Fast Fourier Transform



4 frequency bins/axis

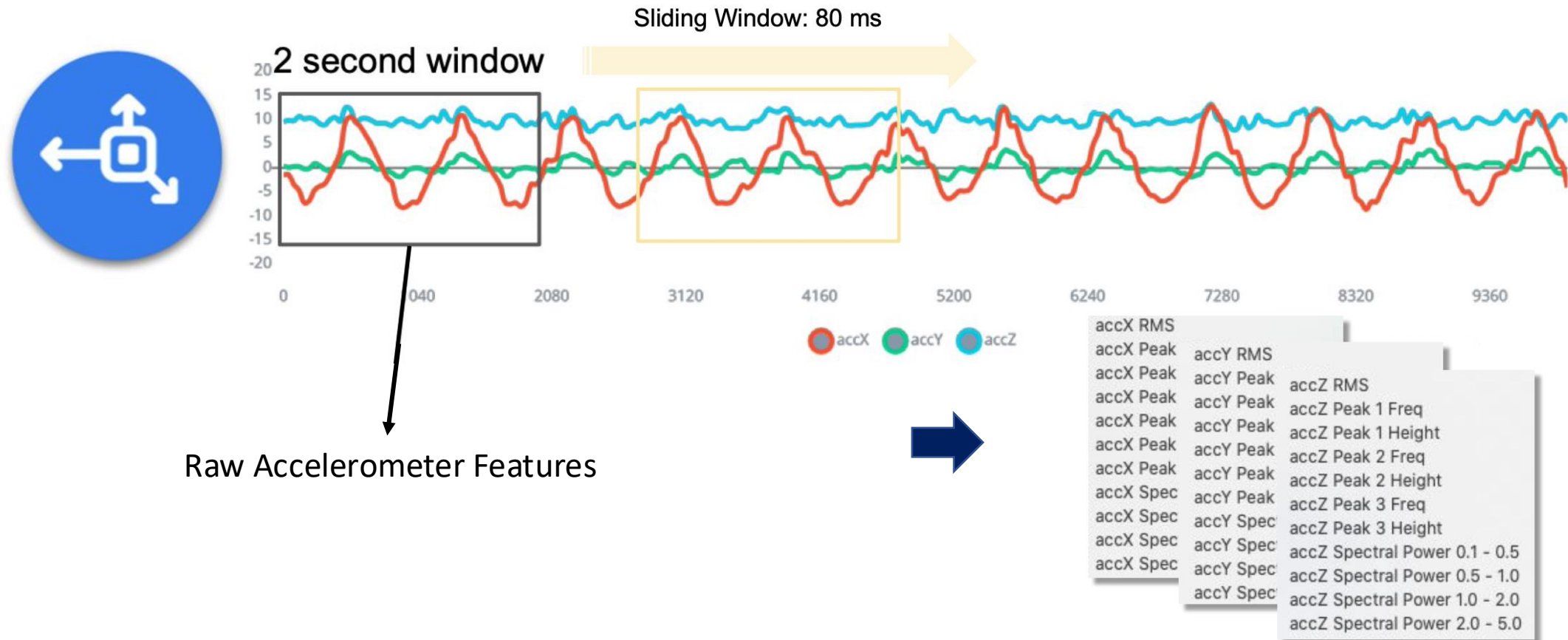


Power Spectral Density

Source: <https://blog.endaq.com/why-the-power-spectral-density-psd-is-the-gold-standard-of-vibration-analysis>

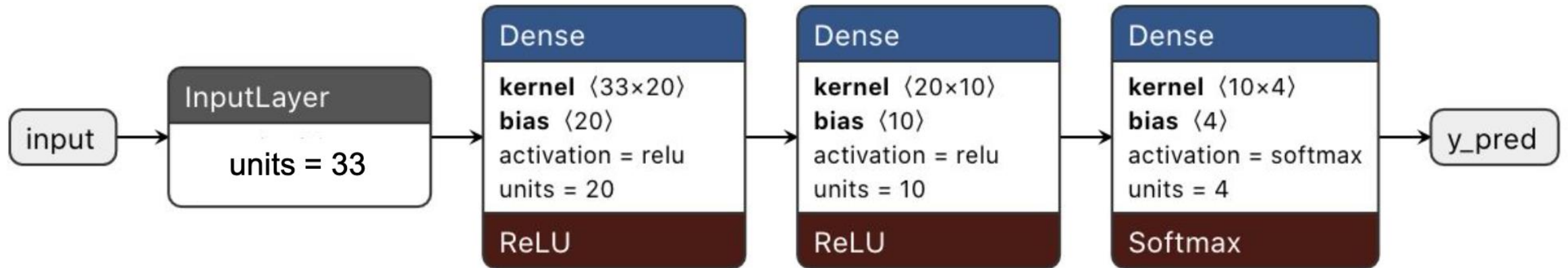
Sliding Window for Accelerometer Data

- For our motion classifier application, we will be using Sliding Window procedure for extracting important features from the raw accelerometer data.

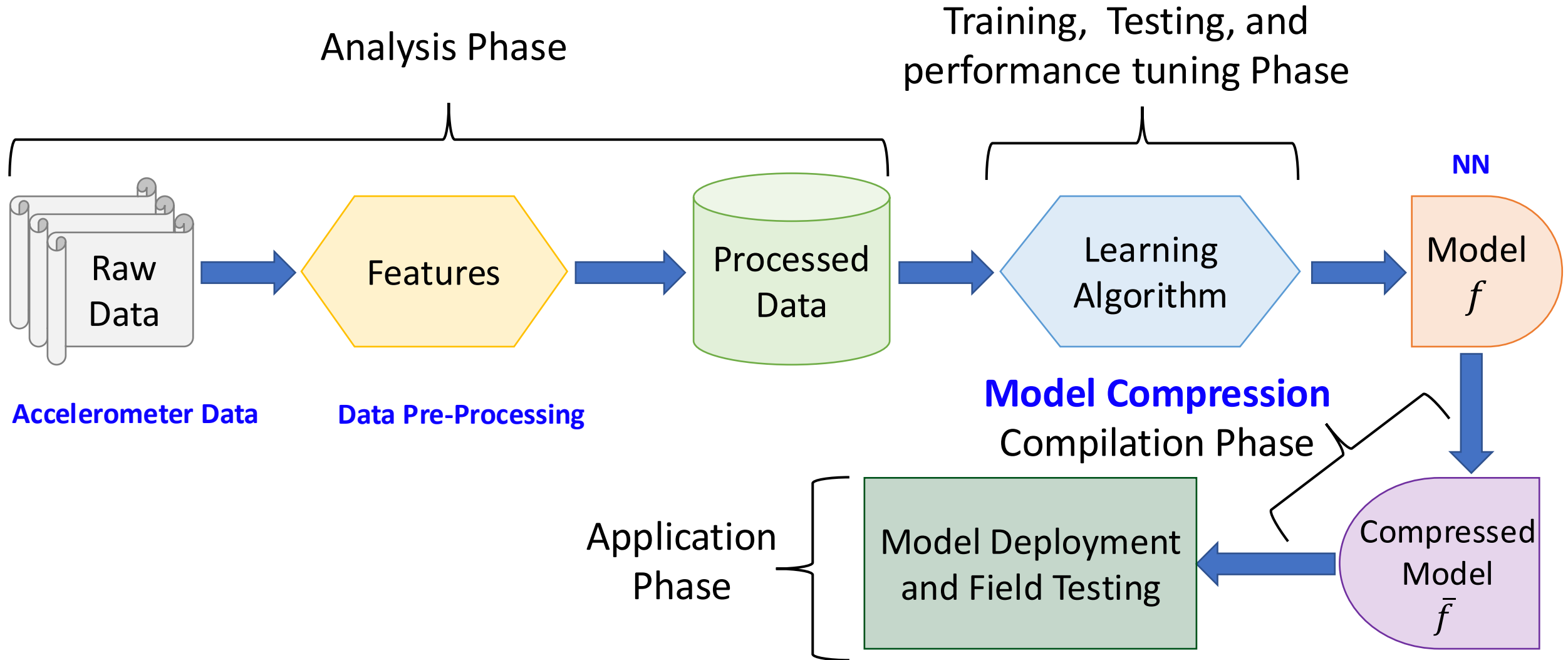


Deep Neural Network (DNN) Classifier

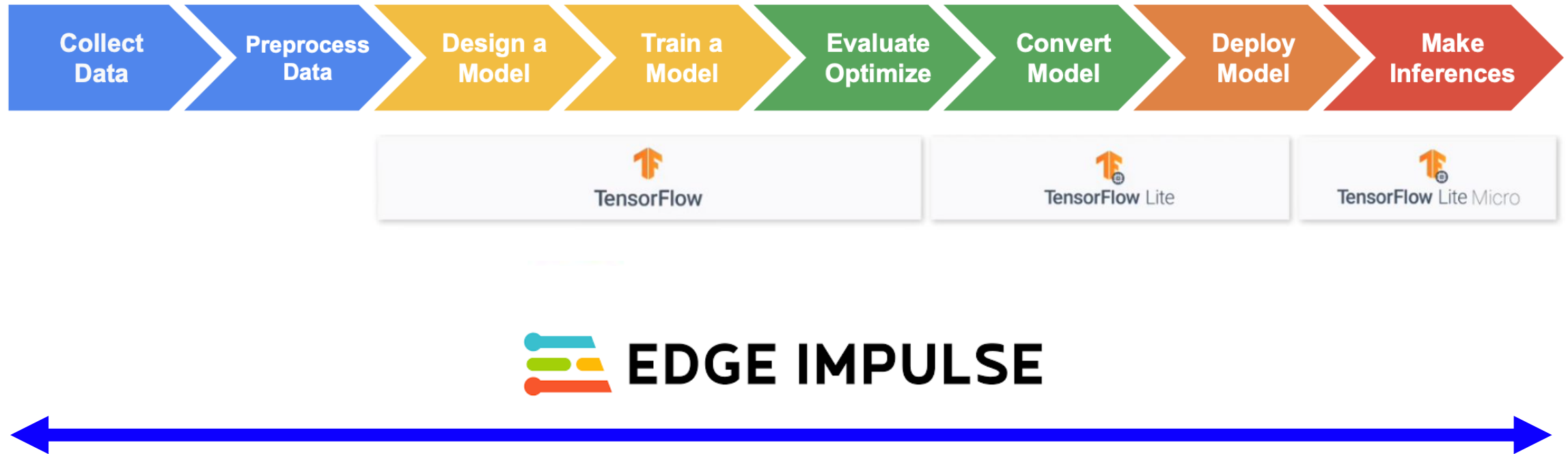
- The DNN classifies transportation data into four classes:
 - lift
 - terrestrial
 - maritime
 - idle



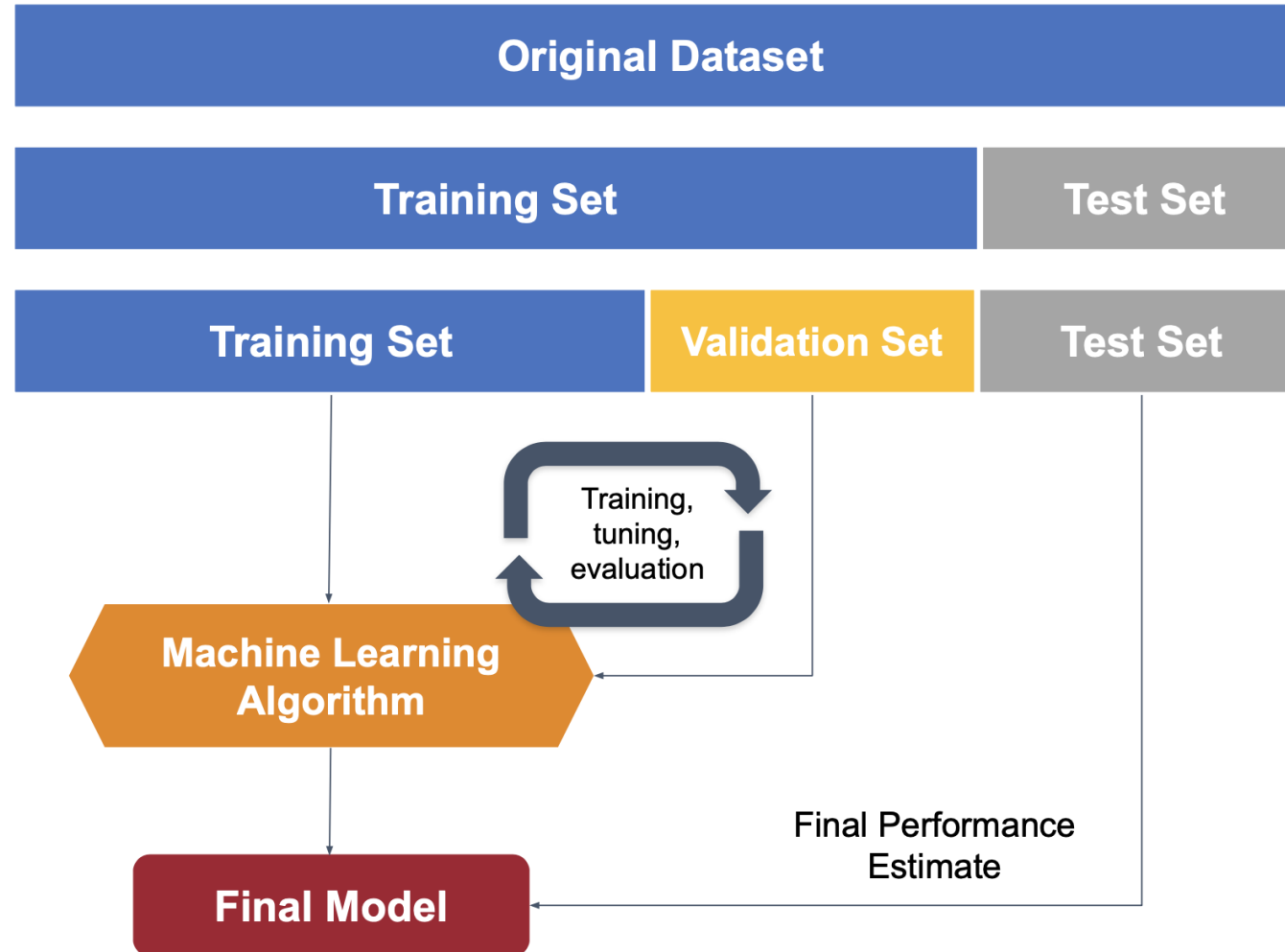
A Schematic View of TinyML Pipeline



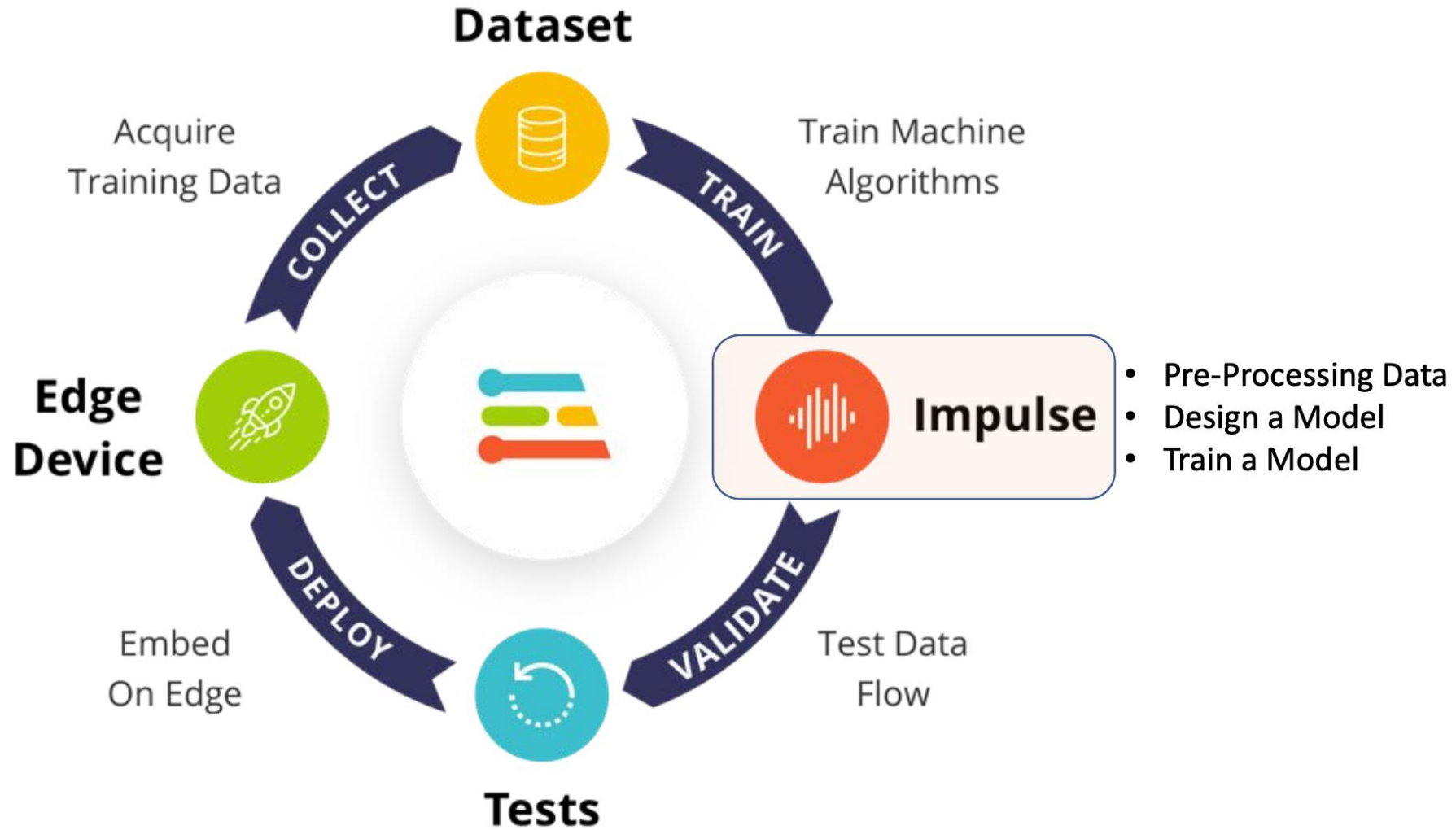
TinyML for Motion Classification



Edge Impulse - Model



Edge Impulse - Model



Today's Lab

- Both Software and Hardware Lab!
- Accelerometer data collection for motion classification
- Computing FFT, Spectrogram, and PSD from time series data
- ML Models for Motion Classifier
- Implement the Model with TinyML Kits

